Experiment Title: File and Directory Management in Linux Terminal

Topic: Creating, Copying, Moving, and Removing Files and Directories

1. Objective

To practice Linux file and directory management commands:

- Create folders and nested paths (`mkdir`, `mkdir -p`).
- Create empty files (`touch`).
- Copy, move, and rename files (`cp`, `mv`).
- Delete files and directories (`rm`, `rm -r`, `rm -rf`).

2. Theoretical Background

Linux provides commands to create, duplicate, move, and delete files directly from the terminal without a GUI.

2.1 Commands Practiced in This Lab

Command	Syntax	Simple Meaning / Purpose	Example
`mkdir`	`mkdir <name>`	Create new directory.	`mkdir docs`
`mkdir -p`	`mkdir -p <path>`	Create parent and child folders together.	`mkdir -p proj/src/utils`
`touch`	`touch <file>`	Create empty file or update timestamp.	`touch notes.txt`
`cp`	`cp <src> <dest>`	Copy file to new location/name.	`cp notes.txt backup.txt`
`mv`	`mv <src> <dest>`	Move or rename file/folder.	`mv old.txt new.txt`
`rm`	`rm <file>`	Delete file permanently.	`rm temp.log`
`rm -r`	`rm -r <folder>`	Delete folder and its contents recursively.	`rm -r old_data`
`rm -rf`	`rm -rf <path>`	Force delete folder and files without prompt.	`rm -rf build`

2.2 Important Rules for File Operations

- **No Trash Bin:** `rm` deletes files immediately and permanently.
- **Parent Flag (`-p`):** Creates missing parent folders automatically.
- **Recursive Flag (`-r`):** Removes non-empty folders and their contents.
- **Force Flag (`-f`):** Ignores errors/prompts, deleting without warning.

3. Equipment / Software Requirements

Item	Details	Item	Details
Operating System	Linux (Ubuntu 24.04 LTS)	**Host Machine**	`firefly`
User Account	`ratul`	**Command Shell**	GNU Bash 5.2 / Zsh
Terminal Emulator	Linux Terminal (Kitty / Ghostty)	**Core Tools**	GNU Coreutils (`mkdir`, `touch`, `cp`, `mv`, `rm`)

4. Procedure / Steps

Steps followed in this experiment:

1. Create demo folder (**`mkdir lab3_demo`**) and navigate inside (**`cd`**).
2. Create folder **`docs`** (**`mkdir`**) and nested path **`proj/src/utils`** (**`mkdir -p`**).
3. Create empty files **`notes.txt`** and **`main.c`** using **`touch`**.
4. Copy **`notes.txt`** to **`notes_copy.txt`** using **`cp`**.
5. Move copy to **`backup/`** and rename to **`notes_v1.txt`** using **`mv`**.
6. Delete file with **`rm`**, and delete directories with **`rm -r`** and **`rm -rf`**.

5. Result / Output Demonstration

All commands were executed under user account `ratul` on host machine `firefly`.

5.1 Session 1: Creating Directories, Nested Trees, Empty Files, and Copying

ratul@firefly: ~/lab3_demo — bash

```
ratul@firefly:~/lab3_demo$ mkdir docs backup
ratul@firefly:~/lab3_demo$ mkdir -p proj/src/utils
ratul@firefly:~/lab3_demo$ touch docs/notes.txt proj/src/main.c
ratul@firefly:~/lab3_demo$ ls docs proj/src
docs: notes.txt        proj/src: main.c  utils
ratul@firefly:~/lab3_demo$ cp docs/notes.txt docs/notes_copy.txt
ratul@firefly:~/lab3_demo$ ls -l docs
total 0
-rw-r--r- 1 ratul users 0 Oct  8 14:10 notes.txt
-rw-r--r- 1 ratul users 0 Oct  8 14:12 notes_copy.txt
ratul@firefly:~/lab3_demo$ _
```

Explanation of Session 1 Results:

- **`mkdir`** and **`mkdir -p`**: Created folders; **`-p`** created the entire path `proj/src/utils` at once.
- **`touch`**: Created empty 0-byte files `notes.txt` and `main.c`.
- **`cp`**: Made an exact copy of `notes.txt` as `notes_copy.txt`.

5.2 Session 2: Moving, Renaming, and Removing Files and Directories

ratul@firefly: ~/lab3_demo — bash

```
ratul@firefly:~/lab3_demo$ mv docs/notes_copy.txt backup/
ratul@firefly:~/lab3_demo$ mv backup/notes_copy.txt backup/notes_v1.txt
ratul@firefly:~/lab3_demo$ ls backup
notes_v1.txt
ratul@firefly:~/lab3_demo$ touch temp.log && rm temp.log
ratul@firefly:~/lab3_demo$ mkdir -p temp_cache/data && touch temp_cache/data/cache.bin
ratul@firefly:~/lab3_demo$ rm -r temp_cache
ratul@firefly:~/lab3_demo$ mkdir -p scratch/a/b && touch scratch/a/test.tmp
ratul@firefly:~/lab3_demo$ rm -rf scratch
ratul@firefly:~/lab3_demo$ ls
backup   docs   proj
ratul@firefly:~/lab3_demo$ _
```

Explanation of Session 2 Results:

- `mv`: Relocated `notes_copy.txt` into `backup/` and renamed to `notes_v1.txt`.
- `rm`: Deleted file `temp.log` permanently.
- `rm -r` and `rm -rf`: Recursively removed `temp_cache` and force-deleted `scratch` without prompt.
- Verified remaining folders with `ls`.

6. Conclusion

Key takeaways from this lab:

1. Created directories with **`mkdir`** and nested paths with **`mkdir -p`**.
2. Created files with **`touch`** and duplicated files with **`cp`**.
3. Moved and renamed files with **`mv`**.
4. Deleted files with **`rm`** and directory trees with **`rm -r`** and **`rm -rf`**.